

The logo consists of two interlocking rings. The left ring is divided into three segments of different shades of brown and tan. The right ring is also divided into three segments, with two shades of brown and one tan segment. The text "Introduction to Google Colab" is centered over the intersection of the two rings.

Introduction to Google Colab



Objectives

- Get to know Google Colab, its purpose and the environment.
- Familiarize the environment of Google Colab.
- Learn the basics of coding in Colab.
- Demonstrate how to program statistical computations in Google Colab.

A stack of papers in light blue, grey, and cream colors is shown on the left side of the slide. Below the papers is a large, textured blue brushstroke graphic that extends towards the bottom right.

What is Google Colab?

- To be precise, Colab is a free Jupyter notebook environment that runs entirely in the cloud.
- Most importantly, it does not require a setup and the notebooks that you create can be simultaneously edited by your team members - just the way you edit documents in Google Docs.
- Colab supports many popular machine learning libraries which can be easily loaded in your notebook.

What Colab Offers you?

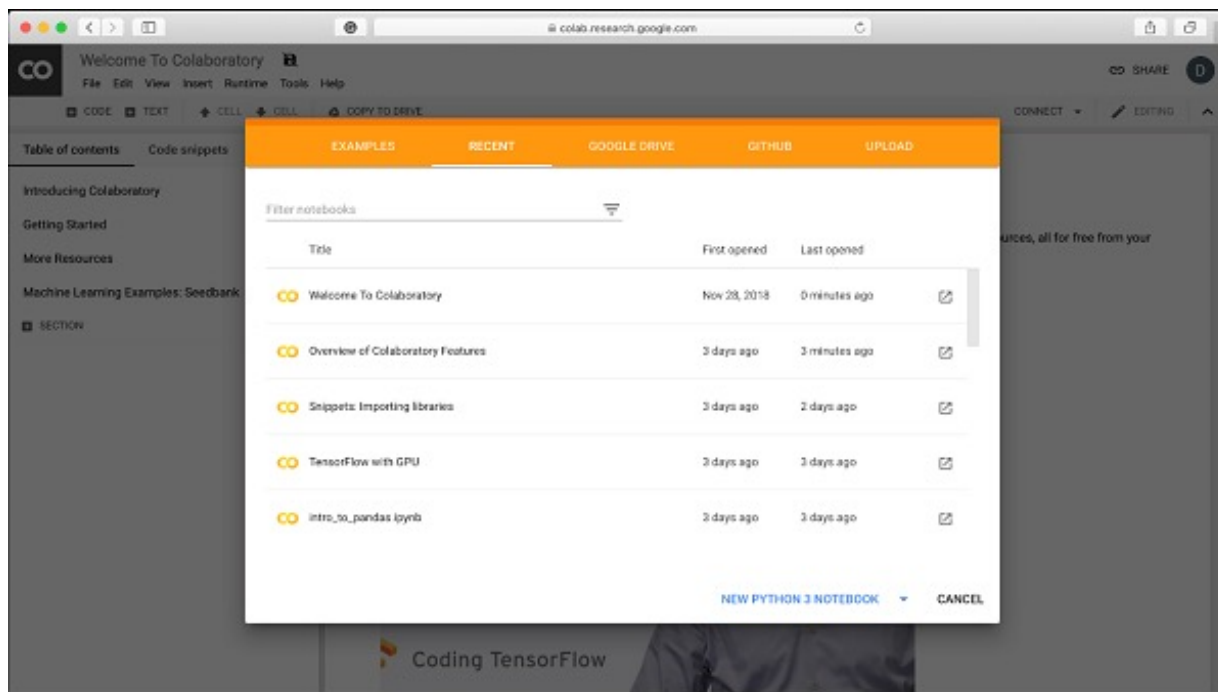
- As a programmer, you can perform the following using Google Colab.
- Write and execute code in Python
- Document your code that supports mathematical equations
- Create/Upload/Share notebooks
- Import/Save notebooks from/to Google Drive
- Import/Publish notebooks from GitHub
- Import external datasets e.g. from Kaggle
- Integrate PyTorch, TensorFlow, Keras, OpenCV



Your First Colab Notebook

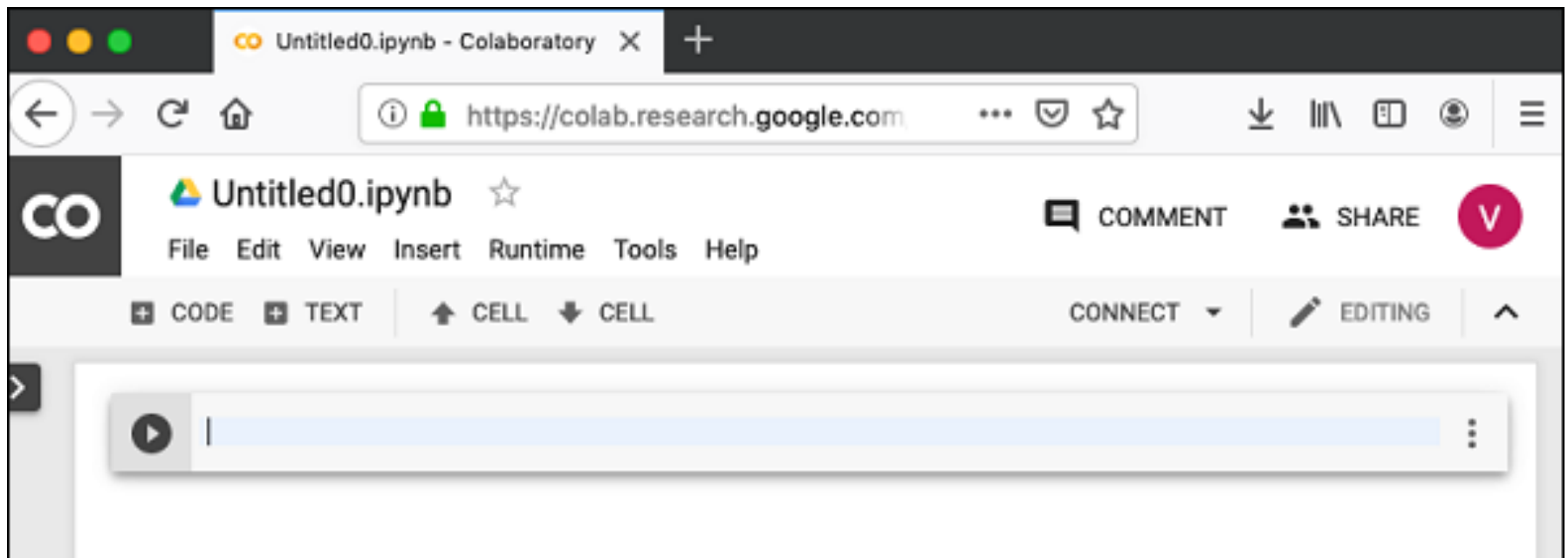
- **Note** – As Colab implicitly uses Google Drive for storing your notebooks, ensure that you are logged in to your Google Drive account before proceeding further.
- **Step 1** – Open the following URL in your browser – <https://colab.research.google.com> Your browser would display the following screen (assuming that you are logged into your Google Drive) –

Your First Colab Notebook



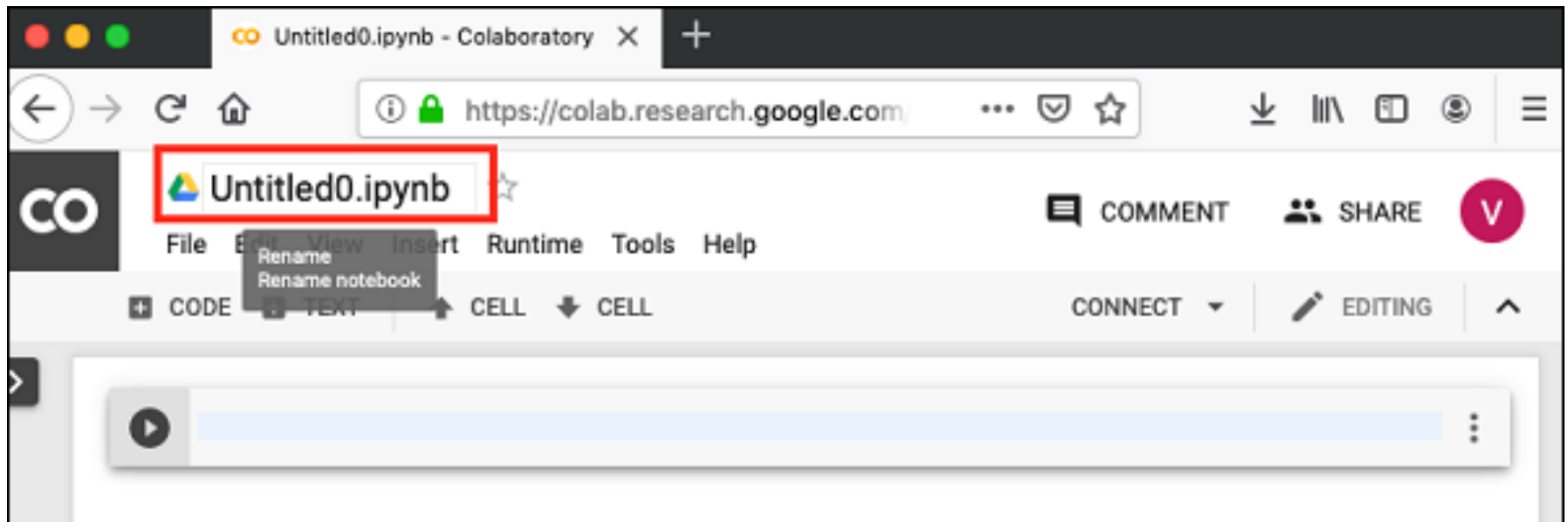
Your First Colab Notebook

- **Step 2** – Click on the **NEW NOTEBOOK** link at the bottom of the screen. A new notebook would open up as shown in the screen below.



Setting Notebook Name

- By default, the notebook uses the naming convention UntitledXX.ipynb. To rename the notebook, click on this name and type in the desired name in the edit box as shown here –



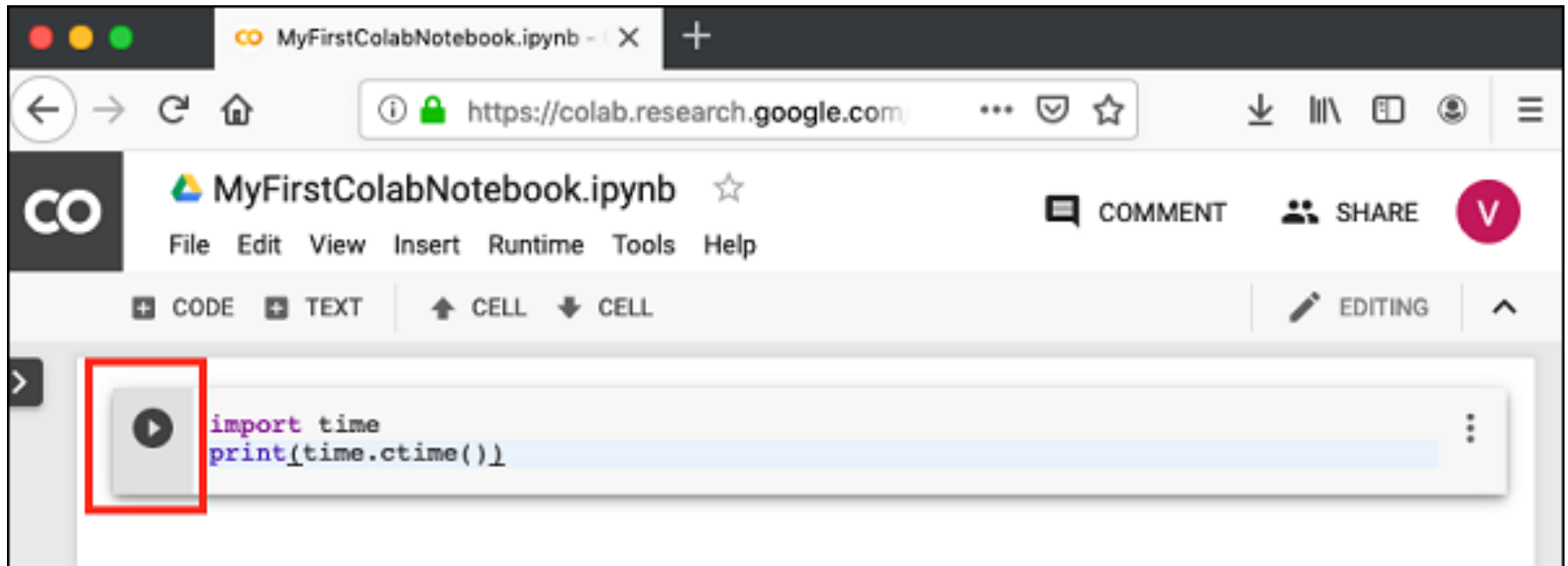
Entering Code

- You will now enter a trivial Python code in the code window and execute it.
- Enter the following two Python statements in the code window
—

```
import time  
print(time.ctime())
```

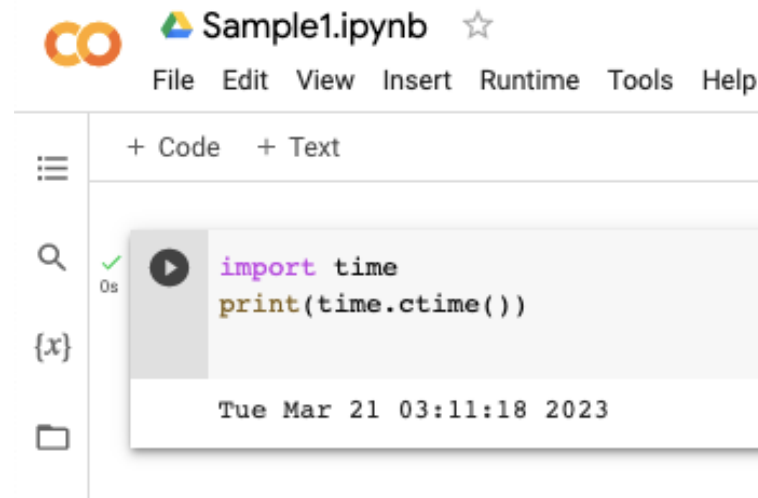
Executing Code

- To execute the code, click on the arrow on the left side of the code window.



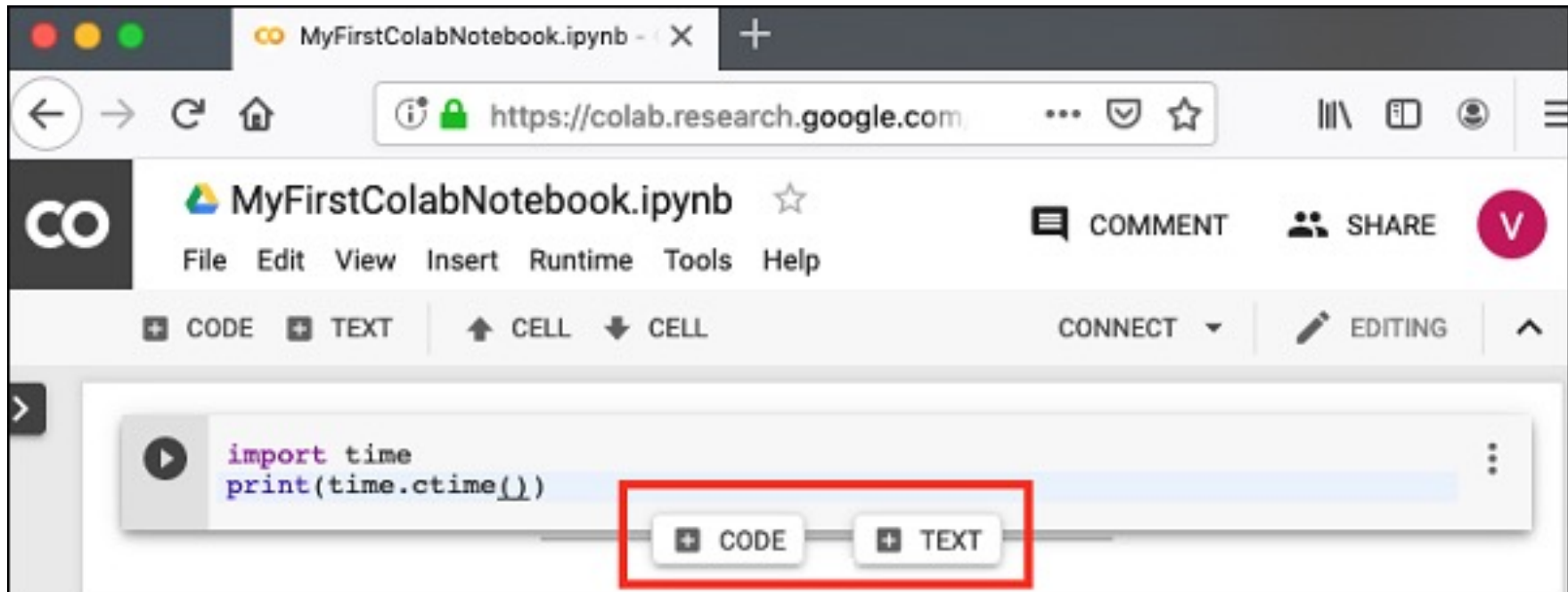
Executing the Code

- After a while, you will see the output underneath the code window, as shown here –
- You can clear the output anytime by clicking the icon on the left side of the output display.

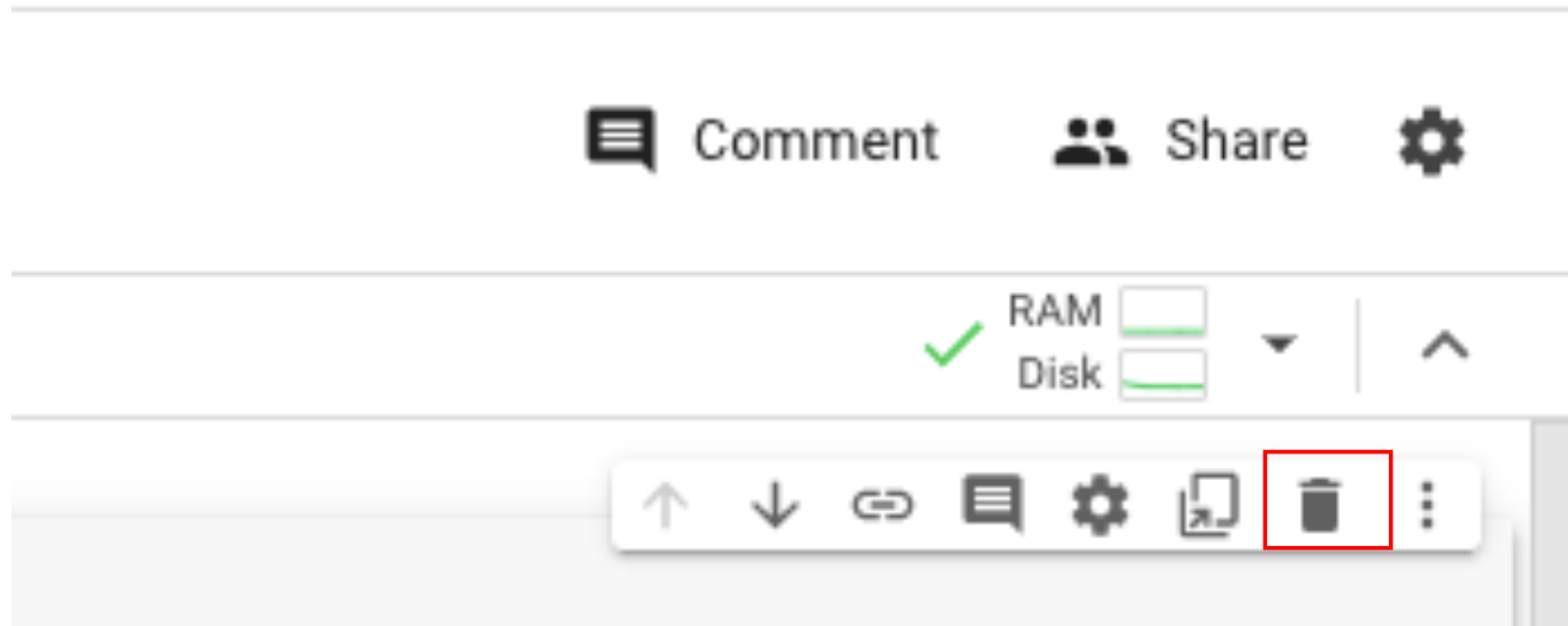


Adding Code Cells

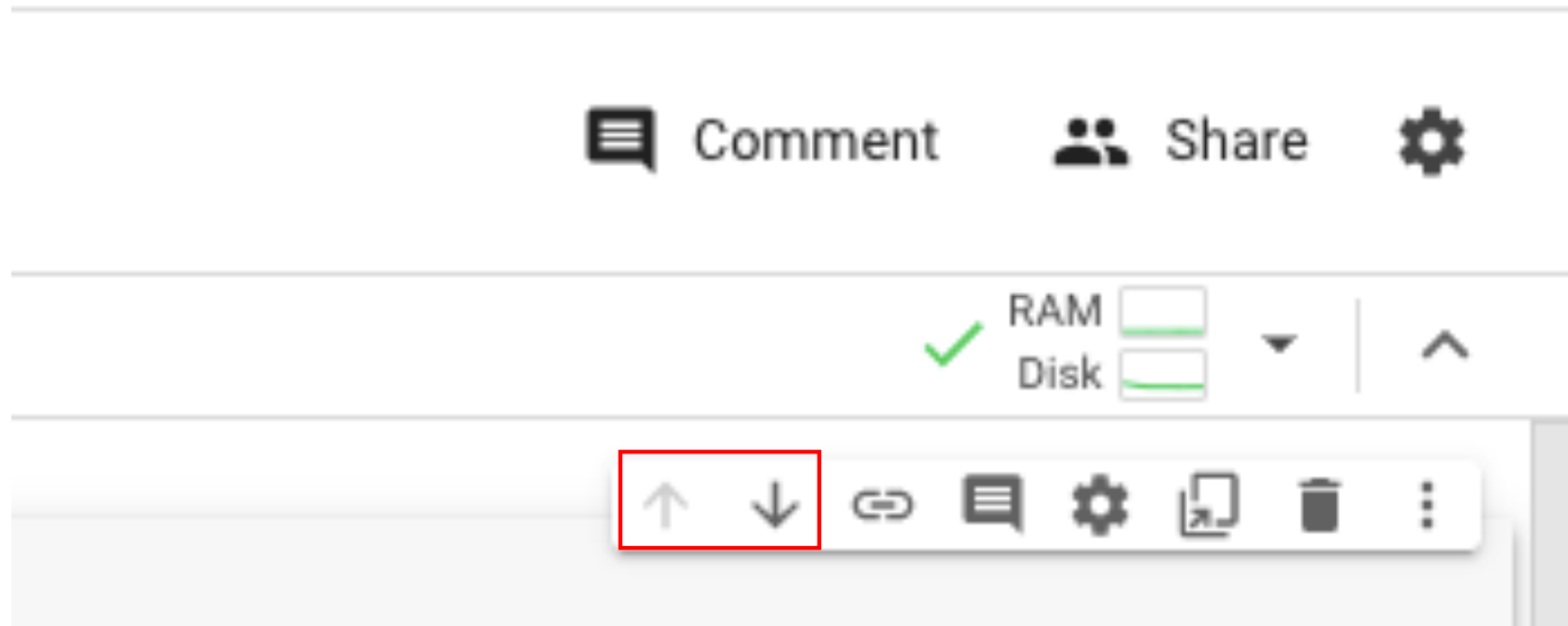
- To add more code to your notebook, select the following options –



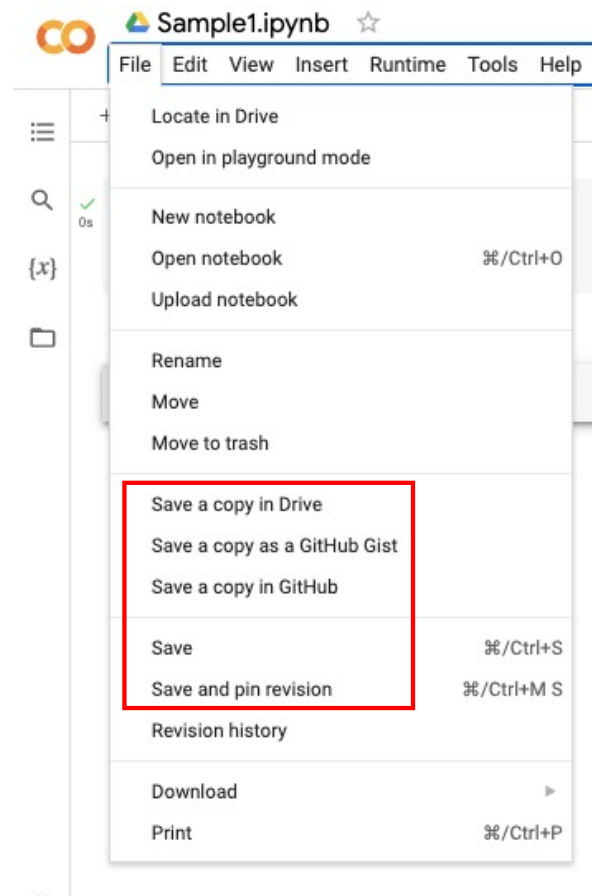
Deleting Cell



Changing Cell Order



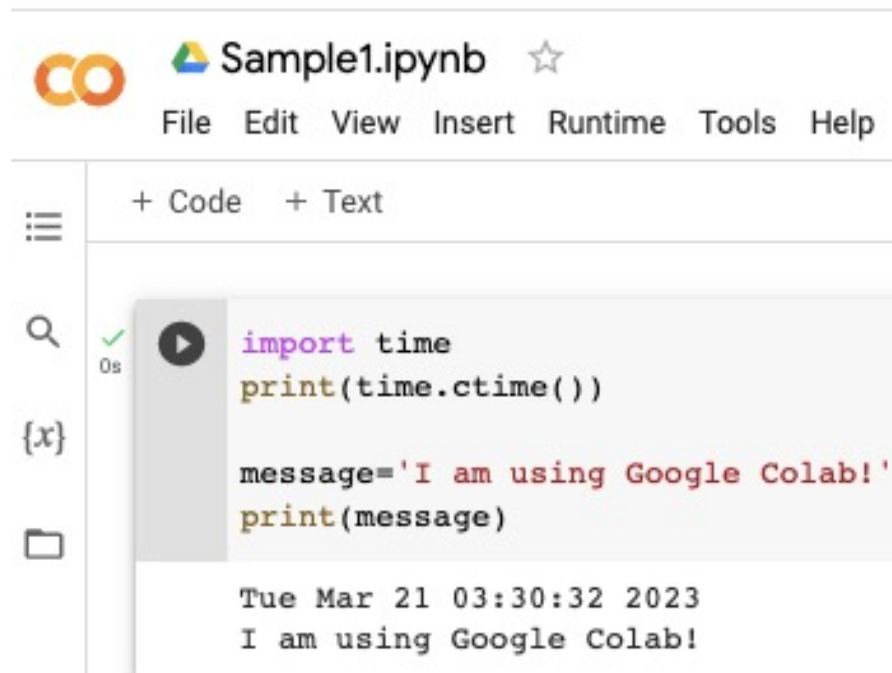
Saving a File





System Commands

Simple Commands



The screenshot shows a Google Colab notebook titled "Sample1.ipynb". The interface includes a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar, there are buttons for "+ Code" and "+ Text". On the left side, there is a sidebar with icons for a menu, search, a variable "{x}", and a file folder. The main area displays a code cell with a play button icon and a green checkmark, indicating successful execution. The code in the cell is:

```
import time
print(time.ctime())

message='I am using Google Colab!'
print(message)
```

The output of the code is displayed below the code cell:

```
Tue Mar 21 03:30:32 2023
I am using Google Colab!
```